

W3WI_110.2 - Verteilte Systeme

Dozent:
Kontakt:
Version:

Prof. Dr. Michael Eichberg
LectureDoc2
michael.eichberg@dhbw.de , Raum 149B
1.0 (23EG/EH)
Developed by M. Eichberg

KI Verwendung:

Mouse Navigation

When you click on the left side of a slide you'll go to the previous slide. When you click on the right half the next step will be done; i.e., the next animation will be started or you'll advance to the next slide.

Touch Support

Slide View

Swipe Gesture - Left and Right

You can swipe left or right to trigger the next animation step or to undo the last step.

Swipe Gesture - Up and Down

You can swipe up or down to go to the previous or next slide.

Pinch Gesture

You will go back to the document view.

Document View

Zoom Gesture

You will change to the slide view and the slide belonging to the section will be shown.

Keyboard Bindings

Key	Action
<i>Navigation</i>	
<space>, →	next step
←	undo last step
<shift> + →	jump to next slide
<shift> + ←	jump to previous slide
<digit(s)> + <enter>	jump to slide number <digit(s)>
b	toggle black screen
r	reset animation progress for current slide
r (5 times in a row)	reset all animations
<i>Presentation</i>	
w	open secondary (linked) window

Kerninhalte gem. MHB

- Terminologie, Konzepte, Architekturen, Anforderungsprofile und Architekturmodelle für verteilte Systeme
- Entwurfs- und Implementierungsansätze
- Vergleich unterschiedlicher Middleware-Konzepte
- Synchron und asynchrone Kommunikation, entfernter Methodenaufruf
- Asynchrone Kommunikation und Messaging-Systeme
- Sicherheit in verteilten Systemen

LectureDoc2

Developed by M. Eichberg

Mouse Navigation

When you click on the left side of a slide you'll go the previous slide. When you click on the right half the next step will be done; i.e., the next animation will be started or you'll advance to the next slide.

Touch Support

Slide View

Swipe Gesture - Left and Right

You can swipe left or right to trigger the next animation step or to undo the last step.

Swipe Gesture - Up and Down

You can swipe up or down to go to the previous or next slide.

Pinch Gesture

You will go back to the document view.

Document View

Zoom Gesture

You will change to the slide view and the slide belonging to the section will be shown.

Keyboard Bindings

Key	Action
<i>Navigation</i>	
<space>, →	next step
←	undo last step
<shift> + →	jump to next slide
<shift> + ←	jump to previous slide
<digit(s)> + <enter>	jump to slide number <digit(s)>
b	toggle black screen
r	reset animation progress for current slide
r (5 times in a row)	reset all animations
<i>Presentation</i>	
w	open secondary (linked) window

Prüfungsleistung - Portfolio

Hintergrund

- das Modul hat 55 VL
- Verteilte Systeme hat 22VL
- Die verteilte Systeme geht mit 50 von 120 Modulpunkten ein

- 2 Bestandteile:

01 Vorträge - max. 15 Punkte - 20 Minuten pro Person

02 Programmieraufgabe - max. 35 Punkte

Achtung!

Personen, die einen Vortrag halten, sollten bei der Programmieraufgabe nicht im selben Team sein.

LectureDoc²

Developed by M. Eichberg

Mouse Navigation

When you click on the left side of a slide you'll go the previous slide. When you click on the right half the next step will be done; i.e., the next animation will be started or you'll advance to the next slide.

Touch Support

Slide View

Swipe Gesture - Left and Right

You can swipe left or right to trigger the next animation step or to undo the last step.

Swipe Gesture - Up and Down

You can swipe up or down to go to the previous or next slide.

Pinch Gesture

You will go back to the document view.

Document View

Zoom Gesture

You will change to the slide view and the slide belonging to the section will be shown.

Keyboard Bindings

Key	Action
<i>Navigation</i>	
<space>, →	next step
←	undo last step
<shift> + →	jump to next slide
<shift> + ←	jump to previous slide
<digit(s)> + <enter>	jump to slide number <digit(s)>
b	toggle black screen
r	reset animation progress for current slide
r (5 times in a row)	reset all animations
<i>Presentation</i>	
w	open secondary (linked) window

Vorträge - Rahmenbedingungen

- Die Präsentationen sollen sich insbesondere mit den Kerninhalten der Vorlesung beschäftigen und insbesondere konzeptioneller Natur sein.

D. h. nach der Darstellung des Anwendungszweckes gilt es die Architektur darzustellen, wie mit Fehlern umgegangen wird, welche Services angeboten werden, welche Garantien/Sicherheitsaspekte umgesetzt werden, wie wird die Skalierbarkeit erreicht, etc.

Keine Werbevorträge!

- Die Präsentationen sind am Abend vor dem ausgemachten Termin hochzuladen in Moodle.

- Die Vorträge dürfen nicht während des Vortrags nicht mehrfach rotieren.

D. h. gefordert ist, dass erst der erste Vortragende seinen Teil vollständig vorträgt, dann der zweite, usw.. Dies ist für die Notenfindung wichtig.

LectureDoc2

Developed by M. Eichberg

Mouse Navigation

When you click on the left side of a slide you'll go the previous slide. When you click on the right half the next step will be done; i.e., the next animation will be started or you'll advance to the next slide.

Touch Support

Slide View

Swipe Gesture - Left and Right

You can swipe left or right to trigger the next animation step or to undo the last step.

Swipe Gesture - Up and Down

You can swipe up or down to go to the previous or next slide.

Pinch Gesture

You will go back to the document view.

Document View

Zoom Gesture

You will change to the slide view and the slide belonging to the section will be shown.

Keyboard Bindings

Key	Action
<i>Navigation</i>	
<space>, →	next step
←	undo last step
<shift> + →	jump to next slide
<shift> + ←	jump to previous slide
<digit(s)> + <enter>	jump to slide number <digit(s)>
b	toggle black screen
r	reset animation progress for current slide
r (5 times in a row)	reset all animations
<i>Presentation</i>	
w	open secondary (linked) window

Vorträge - Themen

Themen, die zu besetzen sind

Thema	Anzahl Stud.
Byzantine Faults 	1 Stud.
LDAP 	2 - 3 Stud.
Backend Subsetting and connection churn 	1 Stud.
Virtualization: Ziel dieser Präsentation ist es, einen Überblick über die verschiedenen Virtualisierungstechnologien zu geben und die Gemeinsamkeiten und Unterschiede zwischen ihnen zu erläutern. Es ist auch möglich, die Unterschiede anhand konkreter Produkte zu demonstrieren. Developed by M. Eichberg	3 - 5 Stud.
Real time system monitoring with eBPF 	2 Stud. (mit Interesse an Linux)
HTTP/3 and QUIC or HTTP over QUIC 	2 Stud.
Themen, die vorgetragen werden können	
Paxos 	2 Stud.
Raft Consensus Algorithm 	2 Stud.
Gossip Protokoll 	2 Stud.
gRPC 	2 Stud.
Web and Distributed Application Testing	4 Stud.
Client und server side Test: alle diskutiert werden	
Neo4J 	2 Stud.

Keyboard Bindings

 connection churn =  Verbindungsabbruch	
Key	Action
Navigation	
<space>, →	next step
←	undo last step
<shift> + →	jump to next slide
<shift> + ←	jump to previous slide
<digit(s)> + <enter>	jump to slide number <digit(s)>
b	toggle black screen
r	reset animation progress for current slide
r (5 times in a row)	reset all animations
Presentation	
w	open secondary (linked) window

Vorträge - Datum

Pro Termin werden Vorträge von insgesamt 100 Minuten Länge gehalten.

VORLÄUFIGE PLANUNG

Date	Topic
9. April	1. (Byzantine faults), 4. (Virtualization), 2. (LDAP), 5. (Monitoring)
16. April	7. (Cosmos), 11. (Testing), 12. (Neo4j), 13. (Docker Swarm vs. Kubernetes)
23. April	8. (Raft), 9. (Gossip), 3. (Connection Churn), 6. (HTTP/3), 10. (gRPC)

Developed by M. Eichberg

Mouse Navigation

When you click on the left side of a slide you'll go the previous slide. When you click on the right half the next step will be done; i.e., the next animation will be started or you'll advance to the next slide.

Touch Support

Slide View

Swipe Gesture - Left and Right

You can swipe left or right to trigger the next animation step or to undo the last step.

Swipe Gesture - Up and Down

You can swipe up or down to go to the previous or next slide.

Pinch Gesture

You will go back to the document view.

Document View

Zoom Gesture

You will change to the slide view and the slide belonging to the section will be shown.

Keyboard Bindings

Key	Action
<i>Navigation</i>	
<space>, →	next step
←	undo last step
<shift> + →	jump to next slide
<shift> + ←	jump to previous slide
<digit(s)> + <enter>	jump to slide number <digit(s)>
b	toggle black screen
r	reset animation progress for current slide
r (5 times in a row)	reset all animations
<i>Presentation</i>	
w	open secondary (linked) window